

Risk I: Choosing Without Knowing

Advanced Microeconomics 1: Economic Decision Making. Lecture 7

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Where we left the question

- Blocks 1–2 priced the shock after it happened.
- But in December 2021 nobody knew. Forecasts said $\sim 2\%$. Households faced a distribution over next year's real income — what is the cost of not knowing?
- Indexation is inflation-insurance (for the employed). Valuing this requires preferences over risk.

Today's route

1. Formalise risk: lotteries, and preferences over them.
2. Two axioms — continuity and independence — and the Expected Utility Theorem.
3. Money lotteries: risk aversion, certainty equivalents, and the Arrow–Pratt measure.

Formalise: lotteries and preferences over them

Modeling risk

Definition · simple lottery

Let $C = \{1, \dots, N\}$ be a finite set of outcomes. A **simple lottery** is a list $L = (p_1, \dots, p_N)$ with $p_n \geq 0$ and $\sum_n p_n = 1$. The set of all simple lotteries is $\mathcal{L}$.

- Outcomes can be consumption bundles or monetary payoffs; probabilities are (for now) objectively known.
- Our inflation lottery, seen from December 2021: two states for 2022 —

$$L^{2022} = \left(\underbrace{0.8}_{\text{normal: } \pi=2\%}, \underbrace{0.2}_{\text{shock: the draw we studied}} \right)$$

Preferences over lotteries

- A preference relation $\succsim$, now on $\mathcal{L}$, assumed **complete** and **transitive**.

- And **continuous**: for any $L, L', L'' \in \mathcal{L}$, the sets

$$\{\alpha \in [0, 1] : \alpha L + (1 - \alpha)L' \succsim L''\} \quad \text{and} \quad \{\alpha \in [0, 1] : L'' \succsim \alpha L + (1 - \alpha)L'\}$$

are closed.

- What continuity buys: if $L \succ L'' \succ L'$, some mixture is *exactly indifferent* to the middle lottery: $\hat{\alpha}L + (1 - \hat{\alpha})L' \sim L''$. No outcome is infinitely good or bad — small probability changes cannot jump the ranking.
- Continuity $\Rightarrow$ a utility representation $U : \mathcal{L} \rightarrow \mathbb{R}$ exists. So far, nothing new.

The independence axiom

Definition · independence

$\succsim$ on $\mathcal{L}$ satisfies **independence** if for all $L, L', L'' \in \mathcal{L}$ and $\alpha \in (0, 1)$:

$$L \succsim L' \iff \alpha L + (1 - \alpha)L'' \succsim \alpha L' + (1 - \alpha)L''.$$

- Mixing both sides with a common third lottery cannot reverse the ranking.
- Why is this *plausible here and false under certainty*? Under certainty, my preference between bundles of goods 1 and 2 may depend on how much of good 3 I consume — goods are consumed *together*. In a lottery, **only one branch is ever realized**: the third lottery is never consumed alongside the others, so it should not interfere.
- NB: "should not" is normative. Whether it *does* describe people is an empirical question.

The Expected Utility Theorem

Theorem · expected utility theorem

If the rational preference relation $\succsim$ on $\mathcal{L}$ satisfies continuity and independence, then it admits a representation of the **expected utility form**: there exist numbers $(u_1, \dots, u_N)$ such that

$$L \succsim L' \iff \sum_{n=1}^N p_n u_n \geq \sum_{n=1}^N p'_n u_n.$$

- Utility of a lottery = *probability-weighted average* of outcome utilities:
 $U(L) = \sum_n p_n u_n$ — the von Neumann–Morgenstern form. Equivalently, U is **linear** in probabilities: $U(\alpha L + (1 - \alpha)L') = \alpha U(L) + (1 - \alpha)U(L')$.
- The cornerstone of modern economics under uncertainty: finance, insurance, game theory, macro all run on it.

Proof → p. 18

Does it survive contact with people?



Maurice Allais, 1911-2010

Does it survive contact with people?

Three prizes: 2.5m, 0.5m, 0. Two choices:

$$1 : L_1 = (0, 1, 0) \text{ vs } L'_1 = (0.10, 0.89, 0.01)$$

$$2 : L_2 = (0, 0.11, 0.89) \text{ vs } L'_2 = (0.10, 0, 0.90)$$

- Most choose L_1 **and** L'_2 . But $L_1 \succ L'_1$ means $u_2 > 0.10 u_1 + 0.89 u_2 + 0.01 u_3$; add $0.89(u_3 - u_2)$ to both sides:

$$0.11 u_2 + 0.89 u_3 > 0.10 u_1 + 0.90 u_3 \iff L_2 \succ L'_2.$$

No vN-M utility generates the modal pattern; the failing axiom is **independence** (both pairs are the same lotteries mixed with a different common third).

- One escape: *regret* — choosing L'_1 and drawing 0 hurts more when certainty was on the table.

Money lotteries and risk aversion

Lotteries over money

- Outcomes are now amounts of money x ; a lottery is a cdf $F : \mathbb{R} \rightarrow [0, 1]$, $F(x) = \Pr(\text{amount} \leq x)$, with density f where it exists.
- The expected utility theorem extends:

$$U(F) = \int u(x) dF(x)$$

where $u(\cdot)$ — increasing, continuous — is the **Bernoulli utility function**: the utility of a *sure* amount. U evaluates lotteries; u evaluates money.

- Everything about attitudes to risk lives in the *shape* of u .

Risk aversion: concavity

Definition · risk aversion

A decision maker is **risk averse** if for every lottery F , the sure amount $\int x dF(x)$ is weakly preferred to F itself: $\int u(x) dF(x) \leq u(\int x dF(x))$.

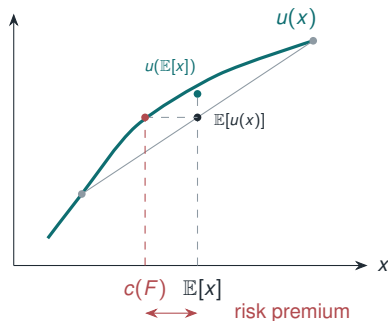
- By **Jensen's inequality**, this holds for all $F \iff u$ is **concave**: diminishing marginal utility of money makes a fair coin-flip of $\pm\text{€}1$ not worth taking.
- Jensen's inequality: $\mathbb{E}[u(x)] \leq u(\mathbb{E}[x])$

Pricing a risk: certainty equivalent and premia

- **Certainty equivalent** $c(F, u)$: the sure amount indifferent to the lottery,

$$u(c(F, u)) = \int u(x) dF(x).$$

- **Risk premium**: $\int x dF - c(F, u) \geq 0$ under risk aversion — what you would pay to swap the lottery for its mean.



Measuring it: Arrow–Pratt

Definition · coefficients of risk aversion

For twice-differentiable u :

$$r_A(x) = -\frac{u''(x)}{u'(x)} \quad (\text{absolute}), \quad r_R(x) = -\frac{x u''(x)}{u'(x)} = x r_A(x) \quad (\text{relative}).$$

- Why divide by u' ? Utilities are unique only up to positive affine transforms; $-u''$ alone is not invariant; r_A is.
- $u(x) = -e^{-ax}$: **CARA**, $r_A \equiv a$. $u(x) = x^{1-r}/(1-r)$: **CRRA**, $r_R \equiv r$.
- u_1 **more risk averse** than $u_2 \iff r_A(\cdot, u_1) \geq r_A(\cdot, u_2) \iff u_1 = \psi(u_2)$, ψ increasing concave.

Where the question stands

- **Machinery built:** rational preferences over lotteries + continuity + independence
⇒ expected utility; risk aversion = concave u ; a risk is priced by its certainty equivalent and premium.
- Allais dents independence.
- **What we cannot yet do:** say when one *distribution* is better or riskier than another, or value the indexation contract.

Appendix

Proof sketch: the Expected Utility Theorem

Assume best and worst lotteries $\bar{L} \succsim L \succsim \underline{L}$ exist (compactness). Three steps:

1. **Calibrate:** by continuity, for each L there is $\alpha_L \in [0, 1]$ with $L \sim \alpha_L \bar{L} + (1 - \alpha_L) \underline{L}$; by independence (monotonicity in the mixing weight), α_L is *unique*. Define $U(L) = \alpha_L$.
2. **Represent:** $L \succsim L' \iff \alpha_L \geq \alpha_{L'}$ — again the mixture-monotonicity that independence delivers.
3. **Linearity:** independence implies $\alpha L + (1 - \alpha) L' \sim \alpha [\alpha_L \bar{L} + (1 - \alpha_L) \underline{L}] + (1 - \alpha) [\alpha_{L'} \bar{L} + (1 - \alpha_{L'}) \underline{L}]$, whose calibration weight is $\alpha \alpha_L + (1 - \alpha) \alpha_{L'}$. So U is linear in probabilities, hence of the form $\sum_n p_n u_n$ with $u_n = U(\delta_n)$.

Independence does all the lifting: it is what makes the calibration weights *add up*.